

Jie Liu

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EDUCATION

- Ph.D. Operation Research, School of Data Science, CUHK(SZ), 2021 – 2025
Honor: *Presidential Fellowship*
Dissertation: *Online Decision Making in Revenue Management: Theory and Applications*
Supervised by Prof. Zizhuo Wang and Prof. Hailun Zhang.
- B.S. Math and Applied Math, School of Science and Engineering, CUHK(SZ), 2017 – 2021
Honor: *First Class Degree, First Class Academic Scholarship (Rank Top 1%)*

ACADEMIC APPOINTMENTS

- 07/2025 – Present Postdoctoral Fellow, Department of IEDA, HKUST
Supervised by Prof. Jiheng Zhang.
- 07/2022 – 01/2023 Research Assistant, Department of IEDA, HKUST

RESEARCH INTERESTS

Game Theory
Pricing and Revenue Management
Stochastic Models
Data-Driven Decision-Making

PUBLICATIONS

- Liu, J., Zhang, H., Niu, B. (2025). The Interplay between Carbon Regulations and Green Technology Innovation: A Comparative Analysis. *International Journal of Production Research*, 1-23. <https://doi.org/10.1080/00207543.2025.2505952>

WORKING PAPERS

- Liu, J., Zhang, H., Zhang, J. When to Match: A Cost-Balancing Principle for Dynamic Markets. Under review at *Management Science*. <https://arxiv.org/pdf/2601.21858>
- Liu, J., Zhang, H., Long, Z. , Zhang, J. Unlocking Inventory Efficiency: The Benefits of Opaque Selling. Working Paper. <https://doi.org/10.2139/ssrn.4817131>
- Liu, J., Chen, Y. J., Wang, Z. Dynamic Price Competition with Uncertain Information. Working Paper.

INVITED TALKS AND CONFERENCE PAPERS

When to Match: A Cost-Balancing Principle for Dynamic Markets.

2026 POMS-HK2026, CUHK(SZ), Jan 2–4.

2025 POMS-CN2025, Hainan, July 11–14.

Unlocking Inventory Efficiency: The Benefits of Opaque Selling.

2025 POMS-HK2025, CUHK, Jan 3–5.

2024 INFORMS Annual Meeting, Seattle, WA. Oct 20–23.

2024 ORSC-DataScience Annual Meeting, Peking, Sept 13–15.

The Interplay between Carbon Regulations and Green Technology Innovation: A Comparative Analysis.

2023 POMS-CN2023, Zhejiang University, July 1–2.

2023 POMS-HK2023, PolyU, Jan 7–8.

2022 POMS-CN2022, XJTU, June 24–27.

COMPETITION AWARDS

2018 **Champion**, Smart City Datathon 2018, Asian Institute of Supply Chains & Logistics, CUHK.

RESEARCH PROJECTS

Impact of information transparency on airline revenue management (with China Southern Airline, 2023–2024)

Served as a core member to explore the impact of information transparency on revenue management via theoretical modeling and numerical simulations.

Authored three industry reports and successfully presented findings to stakeholders, leading to project approval and closure.

INDUSTRY INTERNSHIP

Operation Research Algorithm Intern, Cardinal Operations (05/2024–02/2025)

One of three core algorithm developers for Huanghua Port's Intelligent Integrated Scheduling Platform, China's largest coal port under China Energy Investment Corporation.

Designed and implemented algorithms for unified production scheduling, equipment optimization, and intelligent decision-making, enhancing operational efficiency and green energy dispatching.

Machine Learning Algorithm Research Assistant, Shenzhen Research Institute of Big Data (06/2020–05/2021)

Developed an NLP model based on feature engineering and convolutional neural networks to extract summaries and argumentation information from judicial judgment texts.

TEACHING EXPERIENCE

Teaching Assistant – CUHK(SZ)

MFE5110 Stochastic Process (Fall '23)

DDA4260 Networked Life (Spring '22, Spring '23)

STA2002 Probability and Statistics II (Fall '21)